

ORCA Eval Findings

Evaluation workstream handover — built while Ankit was travelling.

What was built

A three-layer evaluation suite, all powered by Groq (no OpenAI, no RAGAS lib):

- **Layer 1 — retrieval eval** (`run_retrieval_eval.py`): keyword recall/precision over the real `query_for_agentN()` methods. ~12 golden cases.
- **RAGAS-style metrics** (`run_ragas_eval.py`): faithfulness, context_recall, context_precision, answer_relevance — computed with the Groq judge.
- **Layer 2 — LLM-as-judge** (`run_judge_eval.py`): 4 criteria (consistency, hitl_accuracy, scoring_accuracy, classa_safety) over saved decisions.
- **Composite score** (`composite_score.py`): weighted 40/30/20/10 + CI gate.
- **Layer 3 — CI** (`eval_gate.yml`): GitHub Action runs Layer 1 on every push.

Confirmed bug: human briefing does not match the scored decision (Agent 4)

The `hitl_briefing` text names a different option and cost than the `capital_decision` actually selected. The LLM judge flagged 7 of 10 recent runs on the consistency criterion (avg ~2.2/5).

Verified **by hand** on two runs:

SKU	Scored winner (Agent 3)	Briefing says (Agent 4)
SKU00078	Option A, AED 11,623	"Option B, AED 6,953"
SKU00033	Option A, AED 43,424	"Option B, AED 25,815"

Impact: the briefing is what a human reads to approve an order; a mismatch means approving on wrong information. **Likely location:** Agent 4 briefing text in `agents/graph.py`. **Fix:** build the briefing from the winning option in `capital_decision`; keep the judge's `consistency` criterion as a regression guard.

Notes / decisions

- The `ragas` PyPI package could not be installed in this venv — it hard-imports a Vertex AI class incompatible with the repo's LangChain, and a forced install disturbed other deps (pipeline was re-verified working afterward: `from agents.graph import run_pipeline`)

imports cleanly). Decision: compute the same four metrics natively via the Groq judge. The metric definitions are standard; only the library is omitted.

- One Layer-1 case had an over-strict `must_not_contain` removed (the supplier-chain reference table legitimately lists multiple suppliers) — calibration, documented inline.

Next steps (not done yet)

- Calibrate the remaining `[CALIBRATE]` retrieval cases against real wording.
- Wire the RAGAS-style + composite steps into CI (needs a GROQ_API_KEY secret).
- Once the Agent 4 briefing bug is fixed, re-run the judge to confirm consistency scores rise.